FIELD DOCUMENT · MARK-II VAULTS

# VAULT PROTOCOL

EXPERT EYES ONLY

## Vault Security Manual — Revision 1.2

*To the Expert.* Somewhere a vault is ticking. Your partner, the **Operator**, can see it and touch it, but has never read a single page of this manual. You hold the manual but will never see the vault. Listen carefully, ask questions, find the right page and give clear orders. Neither of you can open the vault alone.

### §1 The vault

The vault is a **metal cube**. The Operator turns it with a finger to look at every side. Security **modules** are mounted on the sides, one per side, so the Operator may have to turn the vault to find them all. In Mission 3 the **bottom** carries a module too.

The **front** shows the countdown timer and three strike lights. Each module has a small **status light**: **`RED`** while armed, **`GREEN`** once disarmed.

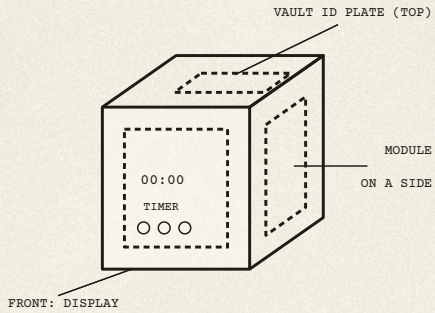



FIG. 1 — MARK-II VAULT, OPERATOR'S VIEW

### §2 Time and strikes

Disarm every module before the timer reaches `00:00`. Every wrong action is a **strike**. The **third strike** sets off the alarm and the attempt is lost.

| Mission | Modules | Time |
|---|---|---|
| 1 · Training Safe | Wires 3–4, Button | `5:00` |
| 2 · Bank Vault | Wires 4–5, Button, Keypad | `4:00` |
| 3 · Central Reserve | Wires 4–5, Button, Keypad, Word Lock | `5:00` |

### §3 The Vault ID — ask for it first

A metal **serial plate on top of the vault** carries the Vault ID: **three letters, a dash and three digits**, e.g. **`KRT-583`**. Several rules depend on it, so have the Operator read it letter by letter before anything else.

| | |
|---|---|
| **Last digit** | the final character: `KRT-583` → 3 |
| **Odd / even** | odd: 1 3 5 7 9 · even: 0 2 4 6 8 |
| **Vowel** | A, E, I, O or U. **Y is not a vowel.** |

### §4 How to read the rules

Rules are checked **from top to bottom**; follow the **first rule that is true** and ignore the rest. The last rule (→) applies when nothing above it is true. Positions are counted **from the top**: wire 1 is the top wire, "first" means nearest the top, "last" nearest the bottom.

### §5 Talking tips

- **Operator:** read the Vault ID letter by letter and describe a module fully before acting.
- **Expert:** repeat the description back before giving an order.
- Use short, exact orders: "cut wire two", "tap it", "hold it", "let go on six".
- When unsure — ask again. A question costs seconds; a strike costs the vault.

SECTION A

# Wires

A panel with **three, four or five wires**. Each wire runs between two sockets; the left socket is numbered (1 is the top wire) and the colour name is printed above the wire.

Only **one** wire is the right one. The Operator cuts a wire by tapping it. A wrong cut is a strike and the wire stays cut; the same rules still apply to the remaining wires.

*Ask:* how many wires? Colours from the top? Last digit of the Vault ID?

Colours used: RED · BLUE · YELLOW · WHITE · GREEN

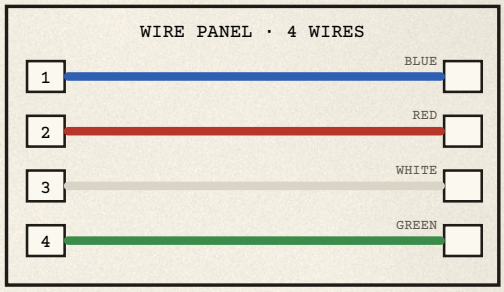



FIG. 2 — WIRE PANEL (USED IN THE FIELD EXAMPLE)

## Cutting instructions — use the table for the number of wires

### 3 WIRES

1. Wire 1 and wire 3 are the **same colour** → **CUT WIRE 2.**
2. There is a **YELLOW** wire → **CUT THE FIRST YELLOW WIRE.**
- → Otherwise → **CUT WIRE 1.**

### 4 WIRES

1. **Two or more RED** wires → **CUT THE LAST RED WIRE.**
2. The Vault ID ends in an **odd** digit **and** there is a **BLUE** wire → **CUT THE FIRST BLUE WIRE.**
3. There is **no GREEN** wire → **CUT WIRE 3.**
- → Otherwise → **CUT WIRE 4.**

### 5 WIRES

1. Wire 5 (the bottom wire) is **GREEN** → **CUT WIRE 4.**
2. There is **exactly one WHITE** wire → **CUT THE WHITE WIRE.**
3. The Vault ID ends in an **even** digit → **CUT WIRE 1.**
- → Otherwise → **CUT WIRE 3.**

### FIELD EXAMPLE · ONE COMPLETE ROUND

**OPERATOR** "The plate on top says K, R, T, dash, five, eight, three."

**EXPERT** "Last digit three, odd. Turn to the wires."

**OPERATOR** "Four wires: one blue, two red, three white, four green."

**EXPERT** "Four-wire table. Two or more red? No, one. Odd digit and a blue wire? Yes. **Cut the first blue wire — wire one.**"

**OPERATOR** taps wire 1.

**VAULT** the wire splits, a chime sounds, the module shows DISARMED and its status light turns green.

| Expert's check, 4-wire table | |
|---|---|
| 1 · 2+ red? | one red → no |
| 2 · odd + blue? | 3 is odd, blue is wire 1 → **yes** |
| 3, 4 | not checked |
| **Order** | **cut wire 1** |

*Had the Operator cut wire 2 instead:* red flash, buzzer, vibration, **strike 1 of 3**. Wire 2 stays cut and wire 1 is still the answer.

SECTION B

# The Override Button

One large round button with a **colour** (red, blue, yellow or white) and a **label**: **OPEN**, **LOCK**, **HOLD** or **RESET**. Below it sit a **light** (off) and a **pin counter** (showing "–").

**Tap**: press and let go at once, before the light comes on. **Hold**: keep pressing. After half a second the light comes on in some colour and the pin counter counts `1, 2 … 9, 1, 2 …`, a new number every 0.8 seconds.

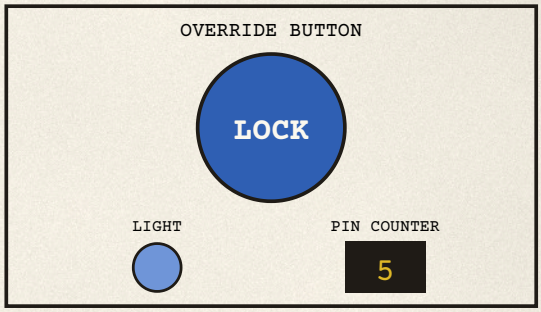



FIG. 3 — BUTTON BEING HELD: LIGHT ON, COUNTER RUNNING

## STEP A — TAP OR HOLD?

1. The label reads **HOLD** → **TAP IT**. Never trust the label.
2. The button is ■ YELLOW → **HOLD IT**.
3. The label reads **LOCK** and the Vault ID ends in an **even** digit → **TAP IT**.
4. The button is □ WHITE → **TAP IT**.

→ Otherwise → **HOLD IT**.

## STEP B — ONLY WHEN HOLDING

Keep holding. When the light comes on, ask its colour, then **let go when the pin counter shows:**

| Light | Release on |
|---|---|
| □ WHITE | **3** |
| ■ BLUE | **6** |
| ■ YELLOW | **2** |
| ■ RED | **8** |

**Strikes:** tapping a button that must be held · holding a button that must be tapped (letting go at any time after the light came on) · letting go on the wrong number. The light picks a new colour on **every** hold, so ask every time.

### FIELD EXAMPLE

**OPERATOR** "Vault ID ends in three. The button is blue and says LOCK."

**EXPERT** "Label HOLD? No. Yellow? No. LOCK with an even digit? Three is odd, so no. White? No. So: **hold it**. What colour is the light?"

**OPERATOR** keeps pressing. "Light is blue. Counter: one… two… three…"

**EXPERT** "Blue means six. **Let go on six.**"

**VAULT** "Released on 6 — correct!" The status light turns green.

SECTION C · MISSIONS 2 AND 3

# Symbol Keypad

Four keys in a 2 × 2 grid, each showing one strange symbol and a small light. The Operator must describe the symbols in plain words — "a horseshoe", "a trident", "a backwards question mark".

The keys must be pressed one at a time in the right order. A correct key lights green and stays accepted; a wrong key is a strike.

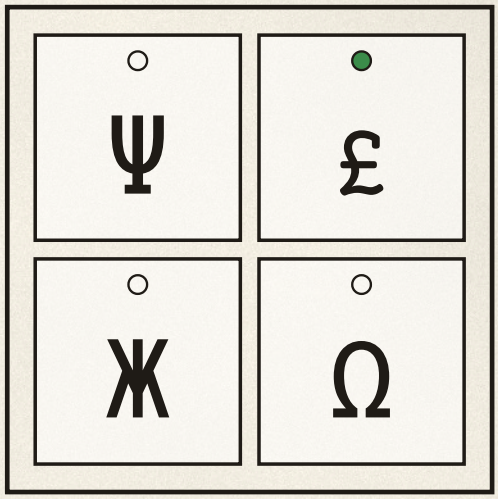



FIG. 4 — KEYPAD, FIRST KEY (£) ACCEPTED

### ORDER OF THE KEYS

1. Find all four symbols in the table below and note their **code numbers**.
2. The Vault ID has **no vowel** → **LOWEST CODE FIRST**, then upwards.

→ The Vault ID has a **vowel** (A E I O U) → **HIGHEST CODE FIRST**, then downwards.

## Symbol codes

| Ψ | £ | Φ | Δ | ¶ | Ж | λ | € |
|---|---|---|---|---|---|---|---|
| 88 | 12 | 72 | 17 | 81 | 30 | 65 | 97 |
| Σ | ¿ | Ω | Ю | π | § | Ξ | ¥ |
| 23 | 05 | 41 | 63 | 09 | 48 | 34 | 56 |

### FIELD EXAMPLE

**OPERATOR** "Vault ID K, R, T. Keys: top-left a trident, top-right a curly L with a line, bottom-left a bug with legs, bottom-right a horseshoe."

**EXPERT** "No vowel — lowest first. Trident 88, curly L 12, bug 30, horseshoe 41. **Press top-right, bottom-left, bottom-right, top-left.**"

**VAULT** each key lights green; after the fourth the keypad is disarmed.

| Vault ID | Order |
|---|---|
| KRT-583<br>no vowel | £ 12 → Ж 30 → Ω 41 → Ψ 88 |
| KAT-583<br>vowel A | Ψ 88 → Ω 41 → Ж 30 → £ 12 |

*Note:* the vault draws the symbols with its own typeface, so their shapes may differ slightly from this table.

SECTION D · MISSION 3 ONLY

# Word Lock

Found on the **bottom of the vault** — the Operator has to turn the vault upside down. A screen shows **one word, or nothing at all**. Below it are **six word buttons** in two columns and three rows, and three **stage lights** on the right.

Many words **sound the same** but are spelled differently. Never guess: the Operator **spells every word letter by letter**.

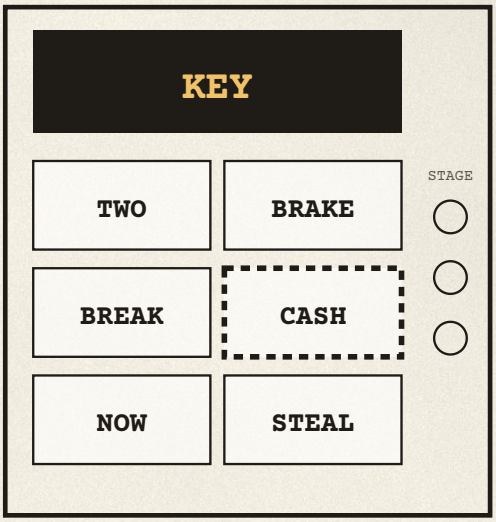



FIG. 5 — WORD LOCK, STAGE 1 (BUTTON TO READ DASHED)

### ONE STAGE

1. Ask for the word on the **screen**. Table 1 says which **button to read**.
2. Ask for the word on **that button** and find its list in Table 2 (next page).
3. Ask for all six button words. Go down the list: the **first word that is on the panel** is the button to press.

→ Right button → the next stage light turns **GREEN** and new words appear. Wrong button → **strike**, new words, same stage. Pass **3 stages**.

## Table 1 — which button to read

Find the screen word; the black square shows the button to read. ZERO is the word Z-E-R-O, OH is O-H, and **(EMPTY)** means the screen shows nothing.

| KEY | KEYS | QUAY | SAFE | SAVE | CODE |
|---|---|---|---|---|---|
| MIDDLE RIGHT | BOTTOM LEFT | TOP RIGHT | BOTTOM RIGHT | TOP LEFT | MIDDLE LEFT |
| **COLD** | **LOCKED** | **LOCK IT** | **ZERO** | **OH** | **(EMPTY)** |
| BOTTOM RIGHT | TOP RIGHT | MIDDLE LEFT | BOTTOM LEFT | MIDDLE RIGHT | TOP LEFT |

Sound-alikes on this module: KEY · KEYS · QUAY | SAFE · SAVE | CODE · COLD | STEEL · STEAL | CASH · CACHE | BREAK · BRAKE | HOLE · WHOLE | TWO · TOO | KNOW · NOW.

SECTION D · CONTINUED

END OF MANUAL

# Word Lock — what to press

## Table 2 — lists

Find the word you read in step 1. Check the list **from left to right**; press the first word that is on one of the six buttons. The word you read is always on the panel, so you will always find one.

| Button word | Look for these words, in this order |
|---|---|
| **STEEL** | BREAK · NOW · WHOLE · STEAL · BRAKE · CASH · TOO · TWO · HOLE · KNOW · CACHE · STEEL |
| **STEAL** | NOW · STEAL · TWO · STEEL · BRAKE · HOLE · KNOW · WHOLE · TOO · BREAK · CACHE · CASH |
| **CASH** | STEAL · BREAK · BRAKE · STEEL · KNOW · CASH · CACHE · TOO · WHOLE · TWO · HOLE · NOW |
| **CACHE** | STEEL · KNOW · STEAL · WHOLE · NOW · TOO · TWO · CACHE · HOLE · BRAKE · CASH · BREAK |
| **BREAK** | BREAK · CASH · WHOLE · TOO · NOW · STEEL · STEAL · BRAKE · TWO · HOLE · KNOW · CACHE |
| **BRAKE** | KNOW · TWO · NOW · CACHE · HOLE · BRAKE · BREAK · CASH · TOO · WHOLE · STEAL · STEEL |
| **HOLE** | BREAK · NOW · STEAL · KNOW · STEEL · CASH · HOLE · TOO · WHOLE · TWO · BRAKE · CACHE |
| **WHOLE** | HOLE · BREAK · STEAL · CACHE · WHOLE · STEEL · BRAKE · KNOW · TOO · TWO · CASH · NOW |
| **TWO** | STEEL · TOO · HOLE · TWO · NOW · KNOW · CACHE · WHOLE · STEAL · BREAK · CASH · BRAKE |
| **TOO** | HOLE · STEAL · KNOW · STEEL · NOW · BRAKE · CASH · TWO · WHOLE · BREAK · TOO · CACHE |
| **KNOW** | CACHE · TWO · BRAKE · CASH · TOO · WHOLE · STEEL · NOW · KNOW · HOLE · BREAK · STEAL |
| **NOW** | TOO · BREAK · KNOW · CACHE · STEAL · STEEL · CASH · NOW · WHOLE · TWO · BRAKE · HOLE |

### FIELD EXAMPLE (FIG. 5)

**OPERATOR** “Screen says KEY — K, E, Y.”
**EXPERT** “KEY: read the **middle right** button.”
**OPERATOR** “C, A, S, H — CASH.”
**EXPERT** “CASH list starts with STEAL. Is there a STEAL — S, T, E, A, L?”
**OPERATOR** “Yes, bottom right.”
**EXPERT** “**Press STEAL.**”
**VAULT** the first stage light turns green and six new words appear.

> Tip: let the Operator read all six buttons once at the start of a stage, from top left to bottom right. Then the Expert can go down the list without asking again.